EDUARDO CAVAZOS QUINTERO

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Education

Massachusetts Institute of Technology **Aug 2025 – May 2029**
Bachelor of Science in Aerospace Engineering GPA: 4.75/5.0 *Cambridge, MA 02139*

- Materials & Structures, Thermodynamics & Propulsion, Fluid Dynamics, Signals & Systems, Principles of Automatic Control, Diff. Equations, Intro to CS & Programming with Python, Statistics & Probability, Physics I/II

South Texas College **Aug 2022 – Jun 2025**
Associate of Interdisciplinary Science with a Major in Mechanical Engineering, GPA: 3.87/4.0 *McAllen, TX 78504*

Skills

Design: CFD/StarCCM, CAD/M, Python, SolidWorks, NX/Teamcenter, ANSYS, VSCode, MATLAB, Thermal Desktop
Hardware & Manufacturing: Machining (Manual Mills, Manual Lathes), Waterjet, Welding, 3D Printing, Prepreg Layup & Composite Manufacturing, Drill Press, Band Saw, Power Tools, Sanding, Compression & Tension Testing

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Technical Projects & Build Team Experience

Aerodynamics Team RE for Rear Wing & Active Aero & Sidepods **Jun 2026 – Present**
MIT Motorsports *N52 Edgerton Center @ MIT*

- Built & validated a STAR-CCM+ radiator model, matching air/coolant-side pressure drops (93 Pa/5.1 kPa), & outlet temps. for heat rejection (2.34kW) & mass flow (0.17kg/s) dropping radiator depth 0.08mm saving (0.5kg) in total
- Direct design meetings for MY27 aero package team, scoping full aero redesigns against FSAE 2027 aero rules moving towards a ClA 4.0 ± 0.2, CdA 1.3 ± 0.1, and 42/58 front/rear aero balance within a 14.22 kg, $4,100 budget
- Trained the full aero team on Star-CCM+ sim workflows, & leading onboarding & training of 7 new members
- Updating Rear Wing and implementing new active aero components as well as sidepods, not used in previous years

Composites Team Member RE **Aug 2025 – May 2026**
MIT Motorsports *N52 Edgerton Center @ MIT*

- Fabricated carbon fiber parts for 4 vehicle components (body panels, wing elements, seat shell, battery enclosure) via prepreg and wet layup, vacuum bagging, and bonding which were all significantly lighter than last year~15%
- Tested 2 carbon fiber coupons in tension and compression on an Instron (MIT D-Lab), measuring strength and stiffness to validate layup schedules against design requirements; supported impact attenuator crush testing
- Machined 7 molds, tooling fixtures, & parts on manual mills, lathes, & waterjet; prepared foam plugs for molds

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Research & Lab Experience

UROP Research Assistant **Sep 2026 – Present**
Wright Brothers Wind Tunnel, MIT Aero Astro: PI: Prof. Mark Drela (XFOIL Creator) *Cambridge, Massachusetts*

- Implementing wall-interference & blockage corrections in MATLAB as well as using XFOIL, improving the accuracy of force and pressure data delivered to industry, research, & student test campaigns
- Lead students & industry projects using the tunnel and develop and implement their model then post process data
- Leading an independent project from CAD through CFD and wind tunnel testing of a scale-model airfoil, comparing predicted vs. measured lift, drag, and pressure distribution across 10~15 angles of attack

UROP Research Assistant **Jun 2026 – Sep 2026**
realtalk@MIT, Center for Constructive Communication *Cambridge, Massachusetts*

- Coded 11 conversation transcripts with a structured qualitative codebook, identifying 33 cross-transcript themes
- Authored 6 analytical memos & thematic summaries turning raw data into structured findings for the research lead
- Built participant-demographic visualization slides for a conference presentation; ran a campaign to 15 participants

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Extracurriculars & Other Jobs

AIAA Executive Board Member — MIT Student Chapter **Sep 2026 – Present**

- Helping organize tech talks, industry panels, & engagement events for the aerospace student community & faculty

Orientation Leader & Associate Advisor — MIT OFY **April 2026 – Present**

- Guiding incoming first-years through orientation programming & supporting their transition into campus life

Makerspace Mentor **Feb 2026 – Aug 2026**

- Trained & advised on 3D printer setup while maintaining hardware, managing inventory, & optimizing models

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Certifications

AE1110x: Introduction to Aeronautical Engineering **Delft University of Technology**
Microsoft Office Specialist: Excel Associate (Excel and Excel 2019) **Microsoft**
Microsoft Office Specialist: Word Associate (Word and Word 2019) **Microsoft**